TEMPORAL STRUCTURAL SECURITY

Threat Drift, Structural Trajectory and Runtime Coherence in Live Computational Systems

A Computational Discipline Derived from

Temporal Structural Computing

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Canonical Foundations

VOIGT, R. F. (2026).
TEMPORAL STRUCTURAL COMPUTING — Foundational Manifesto of a New Computational Category.
Zenodo. DOI: 10.5281/zenodo.20114016

VOIGT, R. F. (2026).
VCore Framework — Temporal Structural Runtime Governance.
Zenodo. DOI: 10.5281/zenodo.18249999

Complementary Structural References

- SSRN Paper 6168911
- SSRN Paper 6397018
- ResearchGate Publication 401969333

ABSTRACT

Modern computational security still operates predominantly under reactive paradigms based on isolated events, signatures, posterior correlations, and explicit threat manifestation. Although effective for known scenarios, these models present a critical structural limitation: they observe events, but do not interpret behavioral trajectories over time.

This work proposes a new computational discipline called **Temporal Structural Security**, derived from the foundational category **Temporal Structural Computing**, introducing an approach based on structural coherence, behavioral persistence, temporal memory, and continuous observability of live computational systems.

The model introduces the concept of **Threat Drift**, defined as the progressive deformation of structural computational coherence before explicit threat manifestation. Unlike traditional architectures, the proposed model observes persistent structural behavior, gradual operational degradation, lateral movement, contextual recurrence, and runtime temporal dynamics.

The operational implementation of this discipline is performed through the **VCore Sentinel** framework, a temporal structural governance system focused on interpreting live computational behavior. Sentinel does not operate merely as a detection mechanism, but as an interpretative layer of computational coherence over time.

This work establishes the conceptual foundations of Temporal Structural Security, proposing a new perspective for observability, governance, and computational interpretation in live runtime environments.

1. INTRODUCTION

Modern computing has become distributed, dynamic, ephemeral, and temporally complex. However, most security architectures still rely on models originally designed for static and deterministic environments.

The dominant logic of contemporary cybersecurity can be summarized in four pillars:

- signature
- event
- correlation
- reaction

Within this paradigm, the system reacts only after explicit threat manifestation.

However, computational threats rarely emerge instantly.

Before rupture, there are:

- small incoherences
- progressive degradation
- unusual persistence
- temporally displaced access
- silent modifications
- lateral structural movement
- gradual behavioral deformation

These phenomena often remain invisible to traditional systems.

The central problem is not lack of data.

The problem is lack of temporal structural interpretation.

From this limitation emerges the discipline proposed in this work:

Temporal Structural Security

A computational approach based on structural interpretation of behavior over time.

2. FROM EVENT OBSERVATION TO TRAJECTORY INTERPRETATION

Traditional architectures observe:

- logs
- events
- connections
- alerts
- signatures

However, live systems generate far more complex phenomena:

- behavioral persistence
- operational coherence
- structural memory
- temporal deformation
- contextual recurrence
- structural pressure
- behavioral drift

This work proposes a paradigm transition:

Traditional paradigm

event → alert

Temporal structural paradigm

trajectory → persistence → coherence → drift → rupture

This transition shifts security from reactive logic to interpretative logic.

3. TEMPORAL STRUCTURAL SECURITY

Temporal Structural Security is defined as:

a computational discipline dedicated to temporal structural observation of live computational behavior.

The discipline is based on five pillars:

3.1 Structural Coherence

Computational systems possess natural behavioral patterns over time.

Threats frequently emerge as degradations of this coherence.

3.2 Behavioral Persistence

Threats should not be interpreted merely as isolated actions, but as recurrent behavioral permanence.

3.3 Structural Memory

The system maintains persistent temporal memory of computational behavior.

This enables:

- historical comparison
- evolutionary analysis
- recurrence identification
- contextual anomaly interpretation

3.4 Trajectory Observability

The discipline observes:

- file paths
- structural flows
- process movement
- runtime relations

- network behavior
- operational deformation

3.5 Computational Drift

Threats initially emerge as subtle displacements of structural coherence.

This phenomenon is defined in this work as:

Threat Drift

4. THREAT DRIFT

Threat Drift represents:

the progressive deformation of structural computational behavior before explicit rupture.

Drift is not an event.

It is displacement.

It may manifest as:

- access outside expected operational windows
- abnormal file recurrence
- persistent connections
- silent lateral movement
- gradual coherence degradation
- contextually incompatible structural changes

Drift represents the pre-rupture stage of computational threats.

This approach fundamentally changes traditional observability logic.

5. COMPUTATIONAL SYSTEMS AS LIVE STRUCTURAL FIELDS

Temporal Structural Security proposes an important ontological transition:

Computational systems cease to be treated merely as collections of isolated events.

They become interpreted as:

Live structural fields observable over time

Within this model:

- processes become trajectories
- files become structural entities
- accesses become behavioral vectors
- persistence becomes structural pressure
- drift becomes operational deformation
- threat becomes coherence rupture

This interpretation enables continuous dynamic observability.

6. VCORE SENTINEL

The VCore Sentinel framework operationally implements the principles of Temporal Structural Security.

Its objective is not merely threat detection.

Its objective is:

Interpretation of live structural computational behavior

Sentinel performs:

- continuous observation
- trajectory analysis
- contextual interpretation
- temporal correlation
- structural governance
- operational narrative generation

7. CONCEPTUAL LAYERS OF SENTINEL

The framework is organized into multiple structural layers.

7.1 Structural Threat Graph

A structural topology model responsible for representing:

- computational relations
- trajectories
- persistence
- contextual connections
- behavioral deformations

7.2 Runtime Coherence

Continuous operational coherence analysis layer.

7.3 Structural Memory

Persistent memory of computational behavior.

7.4 Temporal Drift Analysis

Interpretation of gradual behavioral displacement.

7.5 Narrative Intelligence

Interpretative layer responsible for transforming structural complexity into operational human understanding.

8. SECURITY AS STRUCTURAL GOVERNANCE

In the proposed architecture, security ceases to be exclusively protection.

It becomes:

Temporal structural governance of computational behavior

This enables:

- contextual observability
- operational interpretation
- silent degradation detection
- incompatible persistence identification
- structural trajectory tracing
- operational noise reduction
- dynamic coherence analysis

9. APPLICATIONS

Temporal Structural Security possesses cross-domain applicability.

Corporate environments

- insider threat
- silent exfiltration
- lateral movement
- malicious persistence

Cloud and Data Centers

- structural degradation
- operational drift
- distributed anomalous behavior

Industrial environments

- operational coherence
- structural stability
- persistent behavior analysis

Cyber defense

- dynamic observability
- trajectory analysis
- coherence rupture interpretation

Runtime governance

- computational pressure
- structural deformation
- temporal persistence

10. CONCLUSION

This work proposes the discipline:

Temporal Structural Security

as a direct extension of the category:

Temporal Structural Computing

The discipline introduces a new perspective for computational observability based on:

- structural coherence
- temporal memory
- behavioral persistence
- operational trajectory
- structural drift
- contextual interpretation

The concept of Threat Drift enables threats to be interpreted not merely as explicit events, but as progressive deformations of computational coherence.

The VCore Sentinel framework demonstrates the operational viability of this approach through continuous observation of live computational systems.

Thus, security ceases to operate exclusively on isolated events and begins operating on:

Structural behavior over time

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KEYWORDS

Temporal Structural Security
Temporal Structural Computing
Threat Drift
Runtime Coherence
Structural Threat Graph
Structural Memory
Behavioral Persistence
Computational Trajectory
Runtime Governance
Structural Observability
VCore Sentinel
Dynamic Coherence
Live Computational Systems

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